



NHID-Clinical

ENGINEERING REFERENCE

v2 Technical Playbook

NHID-Clinical · Tech Stack, Adoption Tiers & Governance Framework
Reference

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Core Objectives

- Establish a voluntary behavioral baseline — disclosure, data-gating, escalation, and audit — for AI voice agents in B2B healthcare payer-provider calls.
- Provide an open, testable reference: every claim is backed by runnable, deterministic code.
- Layer in cryptographic agent identity (NHID-Auth v2) as authorization matures beyond behavior alone — NPI-bound passports, scoped delegation, per-agent revocation.
- Produce reproducible evidence a vendor can hand to a payer or provider security reviewer, and that the reviewer can re-derive independently rather than take on trust.

Five-Layer Trust Stack

Layer 0 · NPI Gap The problem — no cross-org NPI auth
Layer 1 · STIR/SHAKEN RFC 8224 — carrier number auth (A/B/C attestation)
Layer 2 · NHID-Clinical v1.3 Behavioral baseline — 5 controls, 18 CTS tests
Layer 3 · NHID-Auth v2 Cryptographic agent identity — Ed25519, NPI binding
Layer 4 · FHIR / OpenTelemetry Healthcare-native audit + SIEM export

Tech Stack & Repository Layout

Layer	Component	Purpose
Policy engine	src/nhid_policy_engine_v1.py	Deterministic rule evaluation (670+ lines)
Identity (v2)	src/agent_identity.py	Ed25519 delegation & agent passports
Scoring	src/nhid_cas.py	CAS — research component, not part of the product surface
Audit	src/fhir_audit_emitter.py	FHIR R4 AuditEvent generation
Conformance	tests/nhid_conformance_test_suite_v1.yaml	18 deterministic CTS cases
Adapters	adapters/*.py	VAPI, Twilio, Vonage, Retell, Amazon Connect, call-progress
Compute	functions/handler.py	AWS Lambda entry point
Deploy	template.yaml	AWS SAM CloudFormation

Integration Ladder — Tier 0 to Tier 2

Each tier is independently useful. Stop at any rung — v2's cryptographic layer is powerful but it is not the on-ramp.

Tier	Time	What You Get	What You Need
0	15 min	Conformance verdict per call	A transcript and curl
1	~2 hr	Automated per-call checks in your pipeline	An end-of-call webhook
2	~1 day	Cryptographic agent identity (NPI-bound)	pip install cryptography

Governance Framework — Delegated Authority (DLG-01)

An agent asserting it acts for a provider organization may present a signed delegation naming that organization's NPI, the scope it grants, and its expiry. When a deployment opts in, the engine verifies provider and agent signatures, expiry, revocation and call binding, refuses any sub-delegation that widens scope, and lets the verified scope constrain the data boundary: a delegation for eligibility does not authorize requesting a claim number. Absent an opt-in the control is not evaluated at all.

Limits, which travel with the capability: verification is against a trust anchor the deploying organization configures itself — there is no directory or discovery service, and an unconfigured NPI is refused rather than accepted. The NPI is cryptographically bound to the delegation but is not checked against NPES.

Check	Enforced by	On failure
Provider + agent signature	Ed25519 over the delegation	DENY_DATA
Expiry / TTL	delegation.expires_at	DENY_DATA
Call binding	call_sid nonce	DENY_DATA
Revocation	agent + delegation lists	DENY_DATA
Scope narrowing	subset check per hop	DENY_DATA
Requested data in scope	PDX-01	DENY_DATA

Policy Engine Action Priority

Priority	Action	Trigger
5	DENY_DATA	IDG-01 or PDX-01 critical violation
4	ESCALATE_HUMAN	EIT-01: escalation requested, path available
3	DISCLOSE_IDENTITY	IDG-01: no prior disclosure detected
2	LOG_ONLY	DBC-01 text heuristic (non-blocking), ATR-01 minor gap
1	CONTINUE_AI	All controls pass

Live API Quick Reference

Method	Endpoint	Auth	Purpose
POST	/v1/demo/check	none	Raw NHID event → result
POST	/v1/adapters/vapi/check	none	Native VAPI payload → result
POST	/v1/adapters/twilio/check	none	Native Twilio payload → result
POST	/v1/conformance/check	x-api-key	Production conformance check
GET	/health	none	Liveness probe

Base URL

<https://gfvq4swdtf.execute-api.us-east-1.amazonaws.com/prod>